Identifying attractors with local PCA

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Attractors can be viewed as submanifolds of phase space where generic solutions congregate in the course of time evolution. Their shape depends on the way the space of solutions is parametrised, but the phenomenon itself is captured by the idea of dimensionality reduction of a set of generic solutions. In some cases one may use Principal Component Analysis (PCA) to identify this, but in general the problem is nonlinear and more refined tools are required. We propose using local PCA as an effective approach to identifying attractors without assuming linearity. We apply this to far from equilibrium attractors which have been identified in models of hydrodynamics describing the dynamics of Quark-Gluon Plasma.

I. INTRODUCTION

Hydrodynamics attractors extending into the far from equilibrium domain [1] were proposed as a possible explanation of why Mueller-Israel-Stewart (MIS) theory [2–4] provides a good account of the dynamics of QGP created in heavy ion collisions [5, 6]. Most of the subsequent activity focused on conformal models with strong symmetry assumptions. It was however clear from the outset that to provide a more detailed physical picture one would need a conceptual framework which kept the essential features without simplifications which were too strong, such as the assumed absence of transverse dynamics, or conformal symmetry. Such a general approach was proposed in Ref. [7] (and was recently sharpened by emphasising the role of the singularity of the dynamical system at the origin of proper time [8]). This proposal considers a suitable representation of the space of solutions, perhaps a course-grained one, as the setting in which attractor behaviour can be studied. The emergence of an attractor can then be identified as the convergence of generic solutions to a submanifold of the full phase space. This phenomenon can be described as the approximate dimensionality reduction of a manifold defined by a set of initial conditions in the course of evolution.

In Ref. [7] it was further suggested that one could profitably employ modern techniques of data analysis (and possibly machine learning) to recognise dimensionality reduction in the above sense. A key idea which provides a concrete realisation of the above notion is to consider sections of phase space at fixed values of the proper time. The set of solutions under consideration is seen as a point cloud on such a slice, and data analysis techniques can be directly applied to it. The specific technique used in Ref. [7] was Principal Component Analysis (PCA), which is an effective way to identify linear subspaces (see e.g. Ref. [**?**]). This was a suitable tool for the task, since the parametrisation of phase space was such that in the hydrodynamic regime solutions aligned along a line. In general it may however be physically interesting to use other parametrisations, or consider dynamical systems where the attractor is not linear [8]. This is the problem we consider here: nonlinearity is generic, and needs to be addressed. If one is to use data-driven methods of analysis, the approach to the hydrodynamic attractor cannot be captured by PCA.

We propose the most straightforward remedy: to apply the PCA analysis locally. This is motivated by the idea that the local tangent space captures the dimensionality around a point. There are various aspects of this approach as stated above that need to be addressed, but as we show below it is possible to develop this notion into a framework which can be applied simply and effectively. This we illustrate by applying it to the simplest cases of MIS theory and the HJSW model [9], where it clearly indicates dimensionality reduction in the space of solutions, and thus the emergence of the hydrodynamic attractor. While the models considered here are conformal, this is only assumed for the sake of simplicity: no advantage is taken of this fact. In particular, generic variables are used to parametrise the phase space of solution: we do not introduce coordinates adapted to conformal symmetry to make our analysis possible (as was done in most work following Ref. [1]).

II. THE CONFORMAL MIS ATTRACTOR IN PHASE SPACE

This section briefly reviews the phase space description of the attractor in conformal MIS theory. More complete accounts can be found in Refs. [7, 8, 10, 11].

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A. MIS theory

Using standard notation, the evolution equations of MIS theory are taken to be

$$u \cdot \nabla \mathcal{E} = -(\mathcal{E} + p)\nabla \cdot u + u^\nu \nabla^\mu \pi_{\mu\nu}, \tag{1a}$$

$$(\mathcal{E} + p) u \cdot \nabla u_\mu = -\Delta_{\mu\nu}\nabla^\nu p - \Delta_{\mu\nu}\nabla_\lambda \pi^{\nu\lambda}, \tag{1b}$$

$$\Delta_{\mu\alpha}\Delta_{\nu\beta} u \cdot \nabla \pi^{\alpha\beta} = -\left(1 + \frac{4}{3}\tau_\pi \nabla \cdot u\right)\pi_{\mu\nu} - 2\eta\sigma_{\mu\nu}, \tag{1c}$$

where ∇_μ is the covariant derivative, $\Delta_{\mu\nu} \equiv g_{\mu\nu} + u_\mu u_\nu$ is the transverse projector, $\sigma_{\mu\nu} = \frac{1}{2}\Delta_{\mu\alpha}\Delta_{\nu\beta}(\nabla^\alpha u^\beta + \nabla^\beta u^\alpha - \frac{2}{3}\Delta^{\alpha\beta}\nabla \cdot u)$ is the shear tensor, η is the shear viscosity and τ_π is the relaxation time. The first two of the above equations express conservation of the energy-momentum tensor, while the last is the MIS equation which determines the evolution of the shear-stress tensor. Throughout this paper we set the bulk viscosity as well as the bulk pressure to zero. In this section we will also adopt equations of state and transport coefficients dictated by conformal invariance:

$$\mathcal{E} = 3p = a\,T^4, \qquad \eta = \frac{4}{3} a\, C_\eta T^3, \qquad \tau_\pi = C_\pi / T, \tag{2}$$

where C_η, C_π and a are dimensionless constants and T is the effective temperature.

With the above assumptions, denoting lab frame time by t and the coordinate along the collision axis by z, the energy-momentum tensor for Bjorken flow can be parameterised in terms of two functions of the proper time $\tau \equiv \sqrt{t^2 - z^2}$: the effective temperature T, and the pressure anisotropy $\mathcal{A} \equiv 9\pi^k_k / 2\mathcal{E}$. The physical meaning of the pressure anisotropy is manifest if one expresses it in terms of eigenvalues of the energy-momentum tensor $(T^\nu_\mu) \equiv \mathrm{diag}(-\mathcal{E}, p_L, p_T, p_T)$ as $\mathcal{A} = (p_T - p_L)/p$, (for details see, e.g. [10, 12]). The equations of motion given in Eqs. (1a) to (1c) can then be written in a form describing evolution in proper time as [11]

$$\tau \partial_\tau \ln T = -\frac{1}{3} + \frac{1}{18}\mathcal{A}, \tag{3a}$$

$$\tau \partial_\tau \mathcal{A} = \frac{8 C_\eta}{C_\pi} - \frac{\tau T}{C_\pi}\mathcal{A} - \frac{2}{9}\mathcal{A}^2. \tag{3b}$$

For the purpose of numerical calculations we adopt the $\mathcal{N} = 4$ SYM values for these parameters:

$$C_\eta = 0.08, \quad C_\pi = 0.2, \tag{4}$$

This system is nonautonomous, since the proper time appears explicitly. It is therefore necessary to specify the value of τ to calculate the state of the system. It is then convenient to regard the proper time as an additional dimension of phase space (see e.g. [13]). The resulting extended phase space is thus 3-dimensional, here parameterised by the variables $(T, \mathcal{A}, \tau)$.

In practice it is most convenient to describe the extended phase space in terms of a set of constant proper time slices at fixed values of τ. The evolutions of a set of solutions in the extended phase space can then be depicted on a set of such slices at various times. Specific solutions are represented by dots on a plane parametrised by $(T(\tau), \mathcal{A}(\tau)$

[**MS:** should we have cloud plots here with the attractor shown]

The appearance of the attractor manifests itself as the phenomenon seen in **??**: a generic set of solutions congregates along a submanifold.

The attractor evolves from slice to slice in the the paramtrisation of phase space adopted here, as t does in any generic parameerisation [8]. Furthermore, even its asymptotic shape is dependent on this. In the parametrisation used in Ref. [7] the attractor at late times became approximately linear, so PCA could be used to identify it.

B. PCA

[**MS:** Please write something briefly here: PCA, eigenvalues of the CM, explained variance]

[**MS:** Maybe say that the standard PCA is not great when usng $T, \mathcal{A}$?]

III. LOCAL PCA

[**MS:** Please write something briefly here: explain the idea: local dimension (instead of explained variance).]

NN heuristics: $K = 2d$ captures the tangent space. Variations lead to small changes in the transition regions, which we expect to be controlled by the number of solutions under consideration.

A. Local dimension

[**MS:** Please write something briefly here: two notions of local dimension: cutoff and PR].
We go with dPR.

B. Application of lPCA to MIS

Simplest test.
T, A variables
A plot grid (good for $d = 2$) and a histogram grid (valid for any d!) .
[**MS:** Insert plot: the panel of cloud plots with color coded local dimensions.]
[**MS:** Insert plot: the panel of histograms with color coded pressure anisotropy.]

IV. THE HJSW MODEL

This section describes a hydrodynamic model in which the phase space of Bjorken flow solutions has one more dimension than in the case of MIS theory. This model was proposed as an attempt to account for the damped oscillatory nature of nonhydrodynamic modes of $\mathcal{N} = 4$ SYM theory [9] (see also Ref. [14]). Due to the higher phase space dimensionality, as well as the added complexity of the solutions, this model provides a more challenging test for data-driven methods.

The equations of motion of this model adapted to the symmetries of Bjorken flow can be wrritten in first order form as:

$$\tau T' = \frac{1}{18}\left(\mathcal{A} - 6\right) T, \tag{5}$$

$$\tau \mathcal{A}' = \mathcal{B}, \tag{6}$$

$$\tau \mathcal{B}' = -\frac{1}{27}\mathcal{A}^3 + \frac{2}{3}(1 - 3\Omega_I \tau T)\left(\frac{2}{9}\mathcal{A}^2 - \mathcal{B}\right) - \mathcal{A}\left(\frac{11}{18}\mathcal{B} + \tau^2 T^2 |\Omega|^2\right) + 8 C_\eta \tau T |\Omega|^2 \tag{7}$$

where the prime denotes the derivative with respect to the proper time τ. [**MS:** I rewrote the third eqn above, please check with your code if I did it correctly!] In this study we set the parameter c_σ appearing in Ref. [9] to zero. The parameter

$$\Omega \equiv \Omega_R + \mathrm{i}\Omega_I \tag{8}$$

is intended to match the least-damped quasinormal mode frequency of the AdS_5 black brane. For the purpose of numerical calculations we take

$$\Omega_R = 9.80005, \quad \Omega_I = 2.87631. \tag{9}$$

Thus, the phase space variables will be taken to be $T, \mathcal{A}, \mathcal{B}$. The physical interpretation of $T, \mathcal{A}$ is the same as for MIS theory, while $\mathcal{B}$ is an auxiliary variable introduced so as to cast the evolution equations of [9] in a first order form. There are other (possibly more elegant) ways to write these equations in first order form, but the choice made above is adequate for our purposes.

[**MS:** Insert plot: the panel of cloud plots with color coded local dimensions.]
[**MS:** Insert plot: the panel of histograms with color coded pressure anisotropy.]

V. SUMMARY AND OUTLOOK

We have put forward a simple method which overcomes the limitations of standard PCA analysis applied to the task of identifying hydrodynamic attractors. These limitations stem from the fact that the attractor manifold need not be linear. To remedy this, we suggest to track the time evolution of the local dimensionality of the point cloud which represents a set of solutions on slices of the extended phase space. We have shown that this captures the dimensionality reduction in two simple systems undergoing Bjorken expansion: MIS theory and the HJSW model.

We have not encountered any serious shortcomings of this approach, although it involves choosing the number of nearest neighbours around each point, which gives rise to some dependence of the results. This dependence can however

be reduced by increasing the number of solutions considered. Another issue is how to define the local dimension in terms of the local PCA decomposition. The essential content is contained in the eigenvalues of the correlation matrix, but there is a choice involved in translating that data into a single number, which would be convenient for identyfying the reduction of dimnsionality. We have considered two natural choices, but in principle other possibilities could also be useful.

The main strength of our approach is its generality: it can be applied to other models of QGP dynamics, including MIS theory with transverse dynamics, ok models of kinetic theory. Furthermore, we take no advantage of conformal symmetry present in our examples, and we could just as easily consider cases without it.

It should also be clear that while our motivation is the study of attractor behaviour in the dynamics of QGP created in ultrarelativistic heavy ion collisions, the data-driven approach to attractors in dynamical systems is of much wider interest.

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